PARK JUNGHO (Kyle) - CV

Backend Engineer for Quantitative Trading and Blockchain Systems

Backend engineer building cross-exchange arbitrage, on-chain trading, and cross-chain bridges in Rust, Go, and TypeScript. Builds systems where new exchanges, chains, and databases plug in as adapters. At Kronon Labs, owned 8-exchange order connectivity, PnL, and delta management in a team-built arbitrage system, plus Solana DEX trading bots. At Mantle, a global company, ran Korea developer relations and built the Korea team's content pipeline.

CONTACT

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- **GitHub**: https://github.com/kyle-park-io · **Portfolio**: https://jungho.dev
- **Quant Career**: https://jungho.dev/quant · **Personal Trading Strategies**: https://jungho.dev/personal-quant
- **Live Quant Trading Dashboard**: https://kyle-quant.xyz · password demo2025

PROFESSIONAL EXPERIENCE

Mantle

Developer Relations Engineer | Apr 2026 - Oct 2026 (7 months)
Global institutional infrastructure for real-world assets on the Ethereum L2 Mantle Network
Tech Stack: TypeScript, Next.js, PostgreSQL, Solidity

- **Owned Korea builder onboarding for The Turing Test Hackathon 2026**: built the Korean builder funnel across the full 6-week registration-to-submission window of Phase 2, in a $120K two-phase global flagship program. An offline co-working session, the Seoul hacker house workshop (100+ builders), a live online workshop, and 3 external partner events. In Mantle's official count, South Korea ranked seventh by country share at 4.3%
- **Ran the Q402 builder workshop on agent payments (Mantle × Quack AI, Aug 2026)**: 42 builders connected Q402 to their own AI agents over MCP and paid on Mantle mainnet from the chat window, with no wallet app, no signing prompt, and no MNT for gas. Built the operations site as the first version of pay-lab: wallet registration, live leaderboard, quest verification, and a QR payment bridge, with badges and quests judged from on-chain data. 488 on-chain payments in three hours
- **Designed, built, and deployed the Korea team's 16-module content operations pipeline**: X and Lark collection, Korean translation, per-channel conversion including press releases, two-stage human review, and Telegram and X delivery. A reviewer who never opens a terminal approves the copy and sends it from the same screen. From July to October 2026 it collected 446 X posts and produced 162 translations and 837 channel versions; reviewers approved 106 of the translations
- **Taught a stablecoin payments class at Inha University (Oct 2026)**: a session of the Digital Assets and Market Infrastructure course for finance students, on how stablecoin payments move and what changes when an AI agent pays. Ran two labs on pay-lab: students made a wallet on their phones, sent testnet USDC to a classmate and read the receipt on a block explorer, then signed a spending limit for an AI agent in claude.ai or Codex and watched a request past the limit get refused by both the lab server and the USDC contract. Covered Korea's pending won-stablecoin bills issue by issue: who may issue, reserves and redemption, and interest. Link to the slides, in Korean: https://pay-lab-inha.vercel.app/slides

- **Wrote the Korean technical documentation set (5 documents)**: Mantle block explorer (beginner and deep-dive editions), a DEX comparative deep dive, an AI agent trading path guide, and an asset-withdrawal guide. The deep-dive edition covers modular architecture, Rollup vs. Validium classification, L2's dual gas structure, and why Mantle sits at L2Beat Stage 0. Link to the documents: https://kyle-park.notion.site/Docs-3675e2b105168061b410fdf801eeaac8
- **Co-hosted the monthly Korean X livestream and presented its technical session (Jul - Sep 2026)**: July's RFQ session set AMM, on-chain CLOB, RFQ, and intent execution side by side and became a standalone X article. August's bridge session covered Lock & Mint against Burn & Mint and Mantle's bridge routes: the canonical bridge, LayerZero OFT, and Chainlink CCIP. September's RPC workshop compared seven Mantle RPC providers and priced three of them per month by app type. The August live post drew 18.0K views, the second-highest of the Korean account's posts over Apr-Sep 2026
- **Campus tour of university blockchain clubs (May 2026)**: presented Mantle's builder programs and hackathon tracks at seven clubs over three weeks: KAIST, Korea University, Sungkyunkwan, Ewha Womans, Ajou, Inha, and Kwangwoon. Ran a Mantle testnet contract deployment lab at Sungkyunkwan and a live build of a stablecoin pair-trading bot at Ewha. The Korea University, Ewha, Sungkyunkwan, and KAIST clubs posted session recaps on their X accounts. In the campus research contest introduced on the tour, a Sungkyunkwan club member won the grand prize and the Kwangwoon club won the most-participating-club award
- **Designed and delivered hands-on technical onboarding curriculum**: Mantle infra setup, Bybit API, Byreal Skills CLI, and MerchantMoe, Agni, and Fluxion integrations
- **Ran the Korean official account**: 279 posts, 246K impressions, and a 1.74% engagement rate over Apr-Sep 2026, 1.7× the global account's 1.03%. Korea-originated campaigns and event announcements took all six highest-reach slots

Kronon Labs Co., Ltd.

Trading System Backend Engineer | Apr 2025 - Nov 2025 (8 months)
Tech Stack: Go (Fiber, Fx, gRPC), Python (Django), PostgreSQL, MongoDB, AWS (ECS, ECR, EC2, Parameter Store, KMS), Docker

- **CEX Cross-Exchange Arbitrage System (Rothchild)**: a Taker-Taker and Maker-Taker arbitrage system built with the team. Its Go strategy engine trades spreads across Binance, Bybit, Bitget, KuCoin, MEXC, Gate.io, and Bithumb spot markets, Bybit against Jupiter, Bithumb and Upbit KRW markets against Bybit, and the perpetual DEXs Paradex, Backpack, Extended, and Rails. Owned the console backend for orders, balances, and PnL: added order replacement and lookup methods to the common exchange interface a teammate started, and built order placement, cancellation, replacement, and lookup, user-data WebSockets for fills, and keepalive and reconnection for 8 exchanges including Gate.io, plus Solana DEX. Built PnL with 1- and 5-minute history and 1h, 4h, 1d, and 7d totals, a delta manager that nets each asset's delta and clears it on the exchange holding the largest balance, and a dashboard tracking PnL and positions in real time. Set up recorded exchange responses to test each integration
- **Solana DEX On-Chain Trading Bots**: bots in Go running three alpha strategies (New Pair Detection, Back From The Dead, Fat LP Analysis), 249 commits across 4 repositories. GoPlus security checks and top-30-holder and fresh-wallet-share entry filters, Jupiter swaps and trigger orders, a price cache with 5-minute and 1-hour volatility, tiered sell rules, and Helius WebSocket monitoring of Pump.fun, Raydium, Orca, and Meteora pools. A Solscan call coordinator built on a lock-free ring buffer and a worker pool queues calls at three priorities and enforces the rate limit in one place. 186 test functions and 25 benchmarks

- **News and KOL Signal Trading Engine**: a futures trading engine built with the team. It ingests private Telegram channels and X list tweets, scores each item's similarity to known events with the average of several OpenAI assistants, and routes signals to Binance and Bybit USDT-futures orders. Built the X list-tweet collector, the private Telegram channel subscription, and the multi-assistant scoring. Evaluated the LLM scoring on a 100-item test list built for it and iterated three versions of the assistant instructions
- **Exchange Listing Sniper**: built with the team, a sniper that catches Upbit and Bithumb listing announcements and sends new symbols over gRPC to order servers in Tokyo and Singapore that place Binance futures, MEXC, and Bybit orders. Ran the earlier listing detector, which watched Upbit and Bithumb notices and ordered on Binance futures, Bybit, MEXC, and OKX, in a dev environment and wrote up seven improvements
- **Trading Metrics Collection & Analytics**: for an altcoin performance chart backend built with the team, which ranks the top 50 altcoins against BTC alongside open-interest-to-market-cap ratios from Velo, built the Bybit 1-minute kline WebSocket collector and kline API and added time ranges and BTC change rates to the performance API. Also designed a batched metrics collector for strategy detectors
- **Prediction Market Monitoring (Totoro)**: with the team, a module that pairs Polymarket and Kalshi NFL and Premier League markets by match and pulls their prices

Medium Inc.

Backend Engineer | Sep 2022 - Feb 2024 (1 year 6 months)
Tech Stack: Go, TypeScript (NestJS), Solidity, Hyperledger Fabric 2.2, Kafka, MySQL, AWS, Docker

- **Cross-Chain Bridge Infrastructure**: built with the team, a multisig-verified bridge linking the company L1 mainnet RIGO (Cosmos Tendermint with EVM), the private consortium chain MDL (a Hyperledger Fabric 2.2.2 fork), and Ethereum. A Kafka event pipeline, 17 bridge chaincodes, and Solidity contracts
- **Security Token Platform (ERC-1400)**: an STO token designed and built as a Go chaincode on Hyperledger Fabric to the ERC-1400 interface: partition-based issue, transfer, and redeem, and token wallets
- **RIGO Chain Communicator**: a Go client sending RIGO transactions over Tendermint RPC, deployed in-house
- **Hyperledger Fabric Network Automation**: Bash scripts that automate Fabric CA enrollment, genesis block and channel transaction generation, orderer, peer, and CouchDB startup, channel creation and joins, anchor peer updates, the chaincode package, install, approve, and commit steps, and the block explorer connection. Seven Docker Compose configurations with different channel, organization, and orderer counts on Fabric 2.2.6 and 2.2.13 images, the largest with 3 organizations and 2 etcdraft orderers, plus four configurations that run the same sequence from Fabric binaries on Linux and macOS without Docker

BF Labs Inc.

Backend Engineer | Aug 2022 - Sep 2022 (2 months)

- **Government Research Project**: built the report intake and lookup REST API server for an IITP (Institute of Information & Communications Technology Planning & Evaluation) funded project with Node.js, Express, and MariaDB

EDUCATION & LANGUAGES

- **Kyung Hee University, Seoul**: Bachelor of Engineering in Mechanical Engineering | Mar 2014 - Aug 2022
- **English**: TOEIC Speaking 150/200, Intermediate High · **Korean**: Native

TECHNICAL SKILLS

- **Specializations**: CEX/DEX Trading Systems, Exchange Connectivity and Order Management (REST, WebSocket, FIX/SBE), PnL and Delta Management, Cross-Exchange and Triangular Arbitrage, DeFi Protocols, Cross-Chain Bridges, Smart Contracts (ERC-4626, ERC-1400), Workflow Automation, Technical Writing
- **Languages**: Rust, Go, Python, TypeScript
- **Blockchain**: Solidity (Foundry, Hardhat), Solana (Rust), Hyperledger Fabric
- **Backend Frameworks**: NestJS, Next.js, Fastify, Django, Gin, Fiber, Fx
- **Infrastructure**: AWS, GCP, Docker, Kubernetes, PostgreSQL, MySQL, Kafka, Redis, Vercel
- **AI Tooling**: Claude Code, Codex, MCP servers, agent skills and trigger evals, agent-in-the-loop pipelines
- **Frontend**: React, SolidJS, Astro

SELECTED PROJECTS

mantle-kr-herald: Content Operations Pipeline for the Mantle Korea Team

Tech Stack: TypeScript, PostgreSQL, Vercel, Vitest · Private repository, available on request

- **Agent-in-the-loop translation**: a local coding agent fills a purpose-built worksheet and a human approves it. Only translation pairs that a human approved on the two official accounts become few-shot examples for the next translation, and glossary candidates pass a tuned rejection threshold before going to human review. The 178-term glossary feeds the translation prompt and a term check
- **Hexagonal structure and the PostgreSQL move**: external APIs, files, and storage sit behind 34 ports. Over the 99 commits that moved the record store from files to PostgreSQL and shipped it serverless on Vercel, no existing domain file changed; the move added 19 adapters and swapped the entry points. A unique index closes the double-publish hazard that appears when serverless drops the file lock
- **Scale and tests**: 1,051 commits from July to October 2026, 16 modules, 400 test files, and 5,115 test cases. Storage adapters are tested against Postgres running in process, not a mock

pay-lab: On-Chain Payments Lab Kit for Workshops and Lectures

Tech Stack: Next.js, PostgreSQL, Privy, viem, MCP, Vercel · Private repository, available on request

- **One kit, two events**: started as the operations site for the Aug 2026 Q402 builder workshop, where 42 builders made 488 on-chain payments, and ran the Oct 2026 Inha University class on the same kit with Ethereum Sepolia and Circle's testnet USDC
- **Per-event config**: redesigned into lab modules (wallet creation, send and receipt, spending-limit signature, agent connection, agent payment) with one config file per event choosing the chain, gas sponsorship, funding, and limits. Chain adapters for Mantle, Mantle Sepolia, and Ethereum Sepolia
- **Gas sponsorship and agent limits**: the lab server submits each student's signed transfer and pays the gas, with per-transfer and per-student caps set in the config. A remote MCP server for the class connects to claude.ai and Codex over OAuth, and the agent's wallet tools spend only within the limit the signed-in student signed

Binance Triangular Arbitrage System

Tech Stack: Rust, FIX Protocol, SBE (Simple Binary Encoding)

- Triangular arbitrage engine that receives order books and sends orders over Binance FIX API sessions in SBE encoding, computing fee-adjusted opportunities on the BTC, ETH, and USDT triangle

Polymarket 15-Minute Market Arbitrage Bot

Tech Stack: Go, Rust

- Arbitrage bot for Polymarket's 15-minute crypto markets that buys both outcomes when Yes plus No costs less than $1. Once both legs fill, each pair is worth $1 at settlement or on a merge; fees and one-leg fills decide whether it pays. Drops the signal when the two prices are more than 29ms apart or the older one is more than 29ms old. Order execution is off by default; every opportunity is logged and sent as a Telegram alert

trustline: Off-Chain Security Scanner for Web3 Projects

Tech Stack: TypeScript, Node.js, GitHub Actions · https://github.com/kyle-park-io/trustline

- Built after leading the response to a breach, a Next.js server taken through React2Shell (CVE-2025-55182). **9 scan modules** check the off-chain stack before deploy for the mistakes that let one breach spread: secrets in build output, management ports open to the internet, over-privileged cloud IAM, account-wide credentials, persistence artifacts, and disabled audit logs
- Published on **npm** (`npx trustline-scanner scan .`) and as a **GitHub Marketplace Action**, with SARIF output for GitHub code scanning. A read-only on-chain command recovers the EtherRAT malware's command-and-control values from Ethereum smart contracts and maps the contracts around them; the findings feed the scanner's IOC set

skills: Agent Skill Marketplace for Claude Code and Codex

Tech Stack: Agent skills, Python, Bash · Private repository, available on request

- Turns repeated engineering decisions into skills, packaged as **7 domain plugins** that each repository enables as needed. One source file generates the manifests for both Claude Code and Codex
- **Trigger evals for 8 skills.** The negative queries are the ones that belong to a neighbouring skill, which is what separates a precise trigger from a loose one
- Stack presets (Postgres backend, Next.js on Vercel, Terraform, data analytics) so a new repository starts with the right skills on, plus tested guard hooks on agent file writes and commands

writing-guides: Instructions for Agent-Written Copy

Tech Stack: Markdown, Node.js, Python · Private repository, available on request

- 62 guides for Korean copy, screen copy, translation, talk scripts, and documents. Every rule is tagged with its evidence: a piece of feedback, a published post, or a measurement
- A pre-commit checker (links and anchors, dashes, evidence tags, index completeness, stock closing phrases, section numbering) and a presentation toolchain that builds the PDF, HTML, speaker script, and slide images from one master file

muster: Local Scheduler Dashboard

Tech Stack: React, Vite, Tailwind CSS, Fastify · https://github.com/kyle-park-io/muster

- One screen for Windows Task Scheduler, systemd timers, cron, and Codex and Claude schedules across Windows, WSL, macOS, and Linux. Bound to 127.0.0.1, with a confirmation step that shows the exact command for every change

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AWARDS & CERTIFICATIONS

- **1st Place, DeFi & Composability Track, Mantle Global Hackathon 2025** (DOOR Protocol) | Feb 2026
  - Also selected for a Mantle Incubation Grant. The hackathon drew 500+ submissions
  - Wrote the 7 core contracts (150 test functions) for senior/junior tranche vaults (ERC-4626), a safety module, withdrawal epochs, and a rate oracle, plus the Next.js dashboard and keeper bot
- **3rd Place, Seoulana Hackathon 2025** (Solana blockchain development competition) | Apr 2025
  - Built the Orca Whirlpool pool, position, and liquidity transactions for the entry, Stream Square
- **Engineer General Machinery** (Human Resources Development Service of Korea) | Dec 2020

LEADERSHIP & RESEARCH

Orakle: KAIST-based Blockchain Research Society

Vice President (8th Cohort) | Mar 2026 - Present · **Team Leader, ODA Team (7th Cohort)** | Sep 2025 - Feb 2026

- Society operations and program direction. Organized the 8th cohort's final research conference (Sep 2026, Hana Bank Lounge, Samseong)
- **Wrote the 7th cohort's five published DeFi research papers** (CC BY 4.0): https://github.com/orakle-7th-sda/conference-pdf
  - Stablecoin depeg survey covering CEX cases from 2018 to Feb 2026 and on-chain cases from 2022 to Feb 2026, by venue, magnitude, cause, and duration
  - Solana searcher/solver strategy design. Six strategy families for real-time depeg arbitrage on Jito bundles, Jupiter v6, and Helius WebSocket
  - Plus a 17-category DeFi ecosystem taxonomy and a cross-category stablecoin map
- **6th Cohort BTCFi Bitcoin Options Service** (May 2025 - Nov 2025): one-way option proof of concept on BitVMX-CPU, a Go backend (60 REST APIs, WebSocket, PostgreSQL), and the BitStriker frontend (React, MutinyNet wallet support)